

习题 2

2.1 选择题:

- (1) ACD
- (2) A
- (3) ABC 或者 AB
- (4) AD
- (5) BC
- (6) A
- (7) AD
- (8) AB
- (9) B
- (10) A

2.2 填空题:

2.3 当 $X=1, Y=0, Z=1$ 和 $X=1, Y=1, Z=0$ 时, 计算 $F = \overline{XY} + YZ$ 。

当 $X=1, Y=0, Z=1$ 时: $F=1$;

当 $X=1, Y=1, Z=0$ 时: $F=0$;

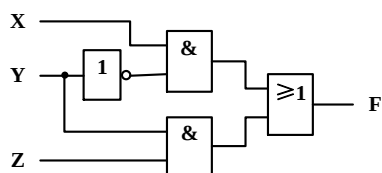
2.4 当 $X=1, Y=0, Z=1$ 和 $X=1, Y=1, Z=0$ 时, 计算 $F = (X+Y)(\overline{Y}+Z)$ 。

当 $X=1, Y=0, Z=1$ 时: $F=1$;

当 $X=1, Y=1, Z=0$ 时: $F=0$;

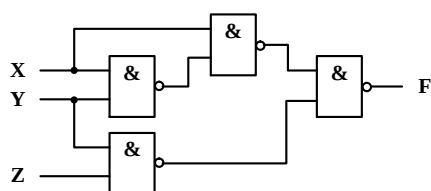
2.7 画出 $F = X\overline{Y} + YZ$ 的电路图, 并将其转换成用与非门表示的电路。请给出 $F = X\overline{Y} + YZ$ 的标准与或式, 并求其等价标准或与式。

(1) $F = X\overline{Y} + YZ$ 电路:



(2) 转换成与非式: $F = X\overline{Y} + YZ = X\overline{X}\overline{Y} + YZ = \overline{\overline{X}\overline{X}\overline{Y}} \cdot \overline{\overline{Y}Z}$

与非门电路:



(3) 标准与或式、标准或与式:

$$F(X, Y, Z) = X\bar{Y}Z + X\bar{Y}\bar{Z} + XYZ + \bar{X}YZ$$

$$= \sum m(3, 4, 5, 7)$$

$$= \prod M(0, 1, 2, 6)$$

$$= (X + Y + Z)(X + Y + \bar{Z})(X + \bar{Y} + Z)(\bar{X} + \bar{Y} + Z)$$

2.9 利用公式证明以下逻辑等式:

$$(1) \quad A \oplus B \oplus C = A \odot B \odot C$$

$$\begin{aligned} A \oplus B \oplus C &= (\bar{A}B + A\bar{B}) \oplus C = (\bar{A}B + A\bar{B})\bar{C} + (\overline{\bar{A}B + A\bar{B}})C \\ &= \bar{A}B\bar{C} + A\bar{B}\bar{C} + \overline{\bar{A}B} \cdot \overline{A\bar{B}} \cdot C = \bar{A}B\bar{C} + A\bar{B}\bar{C} + (A + \bar{B})(\bar{A} + B)C \\ &= \bar{A}B\bar{C} + A\bar{B}\bar{C} + \bar{A}\bar{B}C + ABC = A(\bar{B}\bar{C} + BC) + \bar{A}(\bar{B}\bar{C} + \bar{B}C) \\ &= A(B \odot C) + \bar{A}(B \oplus C) = A(B \odot C) + \bar{A}(\overline{B \odot C}) = A \odot B \odot C \end{aligned}$$

$$(2) \quad \bar{A} \oplus B \oplus C = \overline{A \oplus B \oplus C}$$

$$\text{先证明: } \bar{X} \oplus Y = \overline{X \oplus Y}$$

$$\bar{X} \oplus Y = \bar{X}\bar{Y} + \bar{\bar{X}}Y = \bar{X}\bar{Y} + XY = X \odot Y = \overline{X \oplus Y}$$

则应用到上面的表达式, 有:

$$\bar{A} \oplus B \oplus C = \overline{A \oplus B \oplus C} = \overline{A \oplus B \oplus C}$$

$$(3) \quad \overline{\bar{A}B + B\bar{C} + \bar{A}C} = \bar{A}\bar{B}\bar{C} + ABC$$

$$\overline{\bar{A}B + B\bar{C} + \bar{A}C} = \sum m(1, 2, 3, 4, 5, 6) = \prod M(0, 7) = \sum m(0, 7) = \bar{A}\bar{B}\bar{C} + ABC$$

$$\text{或者 } \overline{\bar{A}B + B\bar{C} + \bar{A}C} = \overline{\bar{A}\bar{B} \cdot \bar{B}\bar{C} \cdot \bar{A}C} = (\bar{A} + B)(\bar{B} + C)(A + \bar{C})$$

$$= (\bar{A}\bar{B} + \bar{A}C + BC)(A + \bar{C}) = ABC + \bar{A}\bar{B}\bar{C}$$

$$(4) \quad \bar{A}\bar{B}\bar{D} + \bar{B}\bar{C}D + \bar{A}D + \bar{A}BC + \bar{A}\bar{B}C\bar{D} = \bar{A}\bar{B} + \bar{A}D + \bar{B}C$$

$$\begin{aligned} \bar{A}\bar{B}\bar{D} + \bar{B}\bar{C}D + \bar{A}D + \bar{A}BC + \bar{A}\bar{B}C\bar{D} &= \bar{B}\bar{D}(A + \bar{A}C) + \bar{B}\bar{C}D + \bar{A}D + \bar{B}C(\bar{A} + A\bar{D}) \\ &= \bar{B}\bar{D}(A + C) + \bar{B}\bar{C}D + \bar{A}D + \bar{B}C\bar{A}\bar{D} = \bar{A}\bar{B}\bar{D} + \bar{B}C\bar{D} + \bar{B}\bar{C}D + \bar{A}D + \bar{B}C \\ &= \bar{A}\bar{B}\bar{D} + \bar{B}\bar{C}D + \bar{A}D + \bar{B}C = \bar{A}\bar{B}\bar{D} + \bar{B}(\bar{C}D + C) + \bar{A}D = \bar{A}\bar{B}\bar{D} + \bar{B}D + \bar{A}D + \bar{B}C \\ &= \bar{B}(\bar{A}\bar{D} + D) + \bar{A}D + \bar{B}C = \bar{A}\bar{B} + \bar{B}D + \bar{A}D + \bar{B}C = \bar{A}\bar{B} + \bar{A}D + \bar{B}C \end{aligned}$$

$$(5) \quad A \oplus B + B \oplus C + C \oplus D = \bar{A}\bar{B} + B\bar{C} + C\bar{D} + \bar{A}D$$

$$\begin{aligned} A \oplus B + B \oplus C + C \oplus D &= \bar{A}\bar{B} + \bar{A}B + B\bar{C} + \bar{B}C + C\bar{D} + \bar{C}D \\ &= \bar{A}\bar{B} + B\bar{C} + C\bar{D} + \bar{A}B + \bar{B}C + \bar{C}D + \bar{A}C + \bar{B}D \\ &= \bar{A}\bar{B} + B\bar{C} + C\bar{D} + \bar{A}B + \bar{B}C + \bar{C}D + \bar{A}C + \bar{B}D + \bar{A}D = \bar{A}\bar{B} + B\bar{C} + C\bar{D} + \bar{A}D \end{aligned}$$

$$(6) \quad \bar{A}\bar{B} + B\bar{C} + C\bar{A} = \bar{A}\bar{B} + \bar{B}C + \bar{C}A$$

$$\begin{aligned} \bar{A}\bar{B} + B\bar{C} + C\bar{A} &= \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + AB\bar{C} + \bar{A}B\bar{C} + \bar{A}BC + \bar{A}\bar{B}C \\ &= (\bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C}) + (\bar{A}B\bar{C} + AB\bar{C}) + (\bar{A}BC + \bar{A}\bar{B}C) = \bar{A}\bar{B} + \bar{B}C + \bar{C}A \end{aligned}$$

$$(7) \quad AB + \bar{A}C + (\bar{B} + \bar{C})D = AB + \bar{A}C + D$$

$$\begin{aligned}
& AB + \overline{AC} + (\overline{B} + \overline{C})D = AB + \overline{AC} + BC + \overline{BC} D = AB + \overline{AC} + BC + D = AB + \overline{AC} + D \\
(8) \quad & BC + D + \overline{D}(\overline{B} + \overline{C})(AD + B) = B + D \\
& BC + D + \overline{D}(\overline{B} + \overline{C})(AD + B) = BC + D + (\overline{B} + \overline{C})(AD + B) \\
& = BC + D + \overline{BC}(AD + B) = BC + D + AD + B = B + D \\
(9) \quad & \overline{A} \overline{C} + \overline{A} \overline{B} + BC + \overline{A} \overline{C} \overline{D} = \overline{A} + BC \\
& \overline{A} \overline{C} + \overline{A} \overline{B} + BC + \overline{A} \overline{C} \overline{D} = \overline{A} \overline{C} + \overline{A} \overline{B} + BC = \overline{A} \overline{BC} + BC = \overline{A} + BC \\
(10) \quad & \overline{A} \oplus \overline{B} = A \oplus B \\
& \overline{A} \oplus \overline{B} = \overline{\overline{A} \overline{B}} + \overline{\overline{A} \overline{B}} = \overline{AB} + \overline{AB} = A \oplus B
\end{aligned}$$

2.12 用代数化简法化简下列逻辑函数：

$$\begin{aligned}
(1) \quad & F = BC + D + (\overline{B} + \overline{C})(AC + B\overline{D}) = BC + D + \overline{A} \overline{B} C + B \overline{C} \overline{D} = BC + D + B \overline{C} + \\
& \overline{A} \overline{B} C = \overline{A} \overline{B} C + B + D = AC + B + D \\
& \text{或者} \\
& F = BC + D + (\overline{B} + \overline{C})(AC + B\overline{D}) = BC + D + \overline{BC}(AC + B\overline{D}) \\
& = BC + D + AC + B\overline{D} = BC + D + AC + B = AC + B + D \\
(2) \quad & F = A + \overline{B\overline{A} + C\overline{D}} = A + \overline{B\overline{A} C\overline{D}} = A \\
(3) \quad & F = \overline{A} \overline{C} + \overline{A} \overline{B} + BC + \overline{A} \overline{C} \overline{D} + AC = \overline{A} \overline{C} + AC + \overline{A} \overline{B} + BC + \overline{A} C = C + \overline{A} + \overline{A} \overline{B} + \\
& BC = C + \overline{A} \\
& \text{或者} \\
& F = \overline{A} \overline{C} + \overline{A} \overline{B} + BC + \overline{A} \overline{C} \overline{D} + AC \\
& = \overline{A} \overline{C} + \overline{A} \overline{B} + BC + AC = \overline{A}(\overline{C} + \overline{B}) + BC + AC \\
& = \overline{A} \overline{BC} + BC + AC = \overline{A} + BC + AC = \overline{A} + BC + C = C + \overline{A} \\
(4) \quad & F = \prod M(5,6,7,10,11,14,15) = \sum m(0,1,2,3,4,8,9,12,13) = \overline{A} \overline{B} \overline{C} \overline{D} + \overline{A} \overline{B} \overline{C} D + \\
& \overline{A} \overline{B} C \overline{D} + \overline{A} \overline{B} C D + \overline{A} B \overline{C} \overline{D} + \overline{A} B \overline{C} D + \overline{A} B C \overline{D} + \overline{A} B C D + (\overline{A} \overline{B} \overline{C} \overline{D} + \\
& \overline{A} \overline{B} \overline{C} D + \overline{A} \overline{B} C \overline{D} + \overline{A} \overline{B} C D) + (\overline{A} \overline{B} \overline{C} \overline{D} + \overline{A} B \overline{C} \overline{D} + \overline{A} B \overline{C} D + \overline{A} B C \overline{D}) + \\
& (\overline{A} B \overline{C} \overline{D} + \overline{A} B \overline{C} D + \overline{A} B C \overline{D} + \overline{A} B C D) = \overline{A} \overline{B} + \overline{C} \overline{D} + \overline{A} C \\
(5) \quad & F = AB + \overline{AC} + \overline{BC} + \overline{A} \overline{B} C D = AB + \overline{AC} + \overline{BC} = AB + \overline{A} \overline{B} C = AB + C \\
(6) \quad & F = AB + [(\overline{A} \oplus C)\overline{B} + \overline{C} D] \overline{B} \overline{D} = AB + (\overline{A} \overline{B} C + \overline{A} \overline{B} \overline{C} + \overline{C} D) \overline{B} \overline{D} \\
& = AB + \overline{A} \overline{B} C \overline{D} + \overline{A} \overline{B} \overline{C} \overline{D} = A(B + \overline{B} C \overline{D}) + \overline{A} \overline{B} \overline{C} \overline{D} = AB + \overline{A} C \overline{D} + \overline{A} \overline{B} \overline{C} \overline{D} \\
(7) \quad & F = (\overline{A} + \overline{B})(B + \overline{C} + \overline{D})(\overline{B} + \overline{C} + D) = (\overline{A} + \overline{B})(B \overline{C} + BD + \overline{B} \overline{C} + \overline{C} + \overline{C} D + \\
& \overline{B} \overline{D} + \overline{C} \overline{D}) = (\overline{A} + \overline{B})(BD + \overline{C} + \overline{B} \overline{D}) = \overline{A} BD + \overline{A} \overline{C} + \overline{A} \overline{B} \overline{D} + \overline{B} \overline{C} + \overline{B} \overline{D} = \overline{A} BD + \\
& \overline{A} \overline{C} + \overline{B} \overline{C} + \overline{B} \overline{D} \\
(8) \quad & F = \overline{(A + B\overline{C})(\overline{A} + D\overline{E})} = \overline{(A + B\overline{C})} + \overline{(\overline{A} + D\overline{E})} = \overline{A} \overline{B} C + \overline{A} D \overline{E} = \overline{A} \overline{B} + \overline{A} C + \\
& \overline{A} D + \overline{A} E
\end{aligned}$$

$$\text{也可以先求对偶: } F' = \overline{A(B + \bar{C}) + \bar{A}(D + \bar{E})} = \overline{AB + A\bar{C} + \bar{A}D + \bar{A}\bar{E}}$$

$$\text{再求对偶的对偶: } F = F'' = \overline{(A + B)(A + \bar{C})(\bar{A} + D)(\bar{A} + \bar{E})}$$

$$= \overline{(A + B) + (A + \bar{C}) + (\bar{A} + D) + (\bar{A} + \bar{E})} = \bar{A}\bar{B} + \bar{A}\bar{C} + \bar{A}\bar{D} + \bar{A}\bar{E}$$

$$(9) \quad F = \overline{(A + B\bar{C})(\bar{A} + \bar{D}E)} = \overline{(A + B\bar{C}) + (\bar{A} + \bar{D}E)} = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{D}\bar{E} = \bar{A}\bar{B} + \bar{A}\bar{C} + \bar{A}\bar{D} + \bar{A}\bar{E}$$

$$\text{也可以先求对偶: } F' = \overline{A(B + \bar{C}) + \bar{A}(\bar{D} + E)} = \overline{AB + A\bar{C} + \bar{A}\bar{D} + \bar{A}E}$$

$$\text{再求对偶的对偶: } F = F'' = \overline{(A + B)(A + \bar{C})(\bar{A} + \bar{D})(\bar{A} + E)}$$

$$= \overline{(A + B) + (A + \bar{C}) + (\bar{A} + \bar{D}) + (\bar{A} + E)} = \bar{A}\bar{B} + \bar{A}\bar{C} + \bar{A}\bar{D} + \bar{A}\bar{E}$$

$$(10) \quad F = A(C + BD)(\bar{A} + BD) + B(\bar{C} + DE) + BC = A(\bar{A}C + \bar{A}BD + BCD + BD) + B\bar{C} + BDE + BC = ABD + B + BDE = B$$

或者

$$F = A(C + BD)(\bar{A} + BD) + B(\bar{C} + DE) + BC$$

$$= A \cdot BD(C + \bar{A}) + B(\bar{C} + DE + C)$$

$$= A \cdot BD(C + \bar{A}) + B = B$$

2.13 求下列逻辑函数的反演函数和对偶函数:

$$(1) \quad \overline{\bar{A}BC + BD(C + AE)}$$

$$(2) \quad A \oplus B + \overline{\bar{A} + D}$$

$$(3) \quad \overline{\bar{A}C} + \overline{\bar{A}B} + BC + \overline{\bar{A}CD} + AC$$

$$(4) \quad F = [(A\bar{B} + C)D + E]B$$

$$(5) \quad F = \overline{\bar{A}B + CD}$$

答:

$$(1) \quad \bar{F} = (A + \overline{\bar{B} + C})[\bar{B} + D + \bar{C}(\bar{A} + E)], \quad F' = (\bar{A} + \bar{B} + \bar{C})[B + \bar{D} + C(A + \bar{E})]$$

$$(2) \quad \bar{F} = (A \odot B)\bar{A}\bar{D}, \quad F' = (A \odot B)\bar{A}\bar{D}$$

$$(3) \quad \bar{F} = (A + C)(A + B)(\bar{B} + \bar{C})(A + C + D)(\bar{A} + \bar{C}),$$

$$F' = (\bar{A} + \bar{C})(\bar{A} + \bar{B})(B + C)(\bar{A} + \bar{C} + \bar{D})(A + C)$$

$$(4) \quad \bar{F} = [(\bar{A} + B)\bar{C} + \bar{D}] \cdot \bar{E} + \bar{B}, \quad F' = [(A + \bar{B})C + D] \cdot E + B$$

$$(5) \quad \bar{F} = \bar{A}\bar{B} + CD, \text{ 或者 } \bar{F} = \overline{(A + B)(\bar{C} + \bar{D})}$$

$$F' = AB + \bar{C}\bar{D} \quad \text{或者} \quad F' = \overline{(\bar{A} + \bar{B})(C + D)}$$

2.22 试用卡诺图化简为最简与或式 (第 (1) ~ (4) 题)、最简或与式 (第 (5) 题):

$$(1) \quad F(A, B, C) = \sum m(0, 1, 2, 4, 5, 7)$$

$$(2) \quad F(A, B, C, D) = \sum m(4, 5, 6, 7, 8, 9, 10, 11, 12, 13)$$

$$(3) \quad F(A, B, C, D) = \sum m(0, 2, 4, 5, 6, 7, 12) + \sum d(8, 10)$$

$$(4) F(A, B, C, D) = \sum m(5, 7, 13, 14) + \sum d(3, 9, 10, 11, 15)$$

$$(5) F(A, B, C, D) = \prod M(0, 1, 4, 7, 9, 10, 13) \cdot \prod D(2, 5, 8, 12, 15)$$

答:

$$(1) F = \bar{B} + AC + \bar{A}\bar{C}$$

BC		A			
		00	01	11	10
A	0	1	1	0	1
	1	1	1	1	0

$$(2) F = \bar{A}B + A\bar{B} + B\bar{C}$$

CD		AB			
		00	01	11	10
AB	00	0	0	0	0
	01	1	1	1	1
	11	1	1	0	0
	10	1	1	1	1

(3)

CD		AB			
		00	01	11	10
AB	00	1	0	0	1
	01	1	1	1	1
	11	1	0	0	0
	10	d	0	0	d

$$F = \bar{A}B + \bar{C}\bar{D} + \bar{A}\bar{D}$$

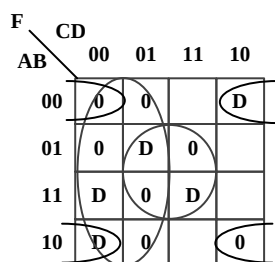
CD		AB			
		00	01	11	10
AB	00	1	0	0	1
	01	1	1	1	1
	11	1	0	0	0
	10	d	0	0	d

$$F = \bar{A}B + \bar{C}\bar{D} + \bar{B}\bar{D}$$

$$(4) F = CD + BD + AC$$

CD		AB			
		00	01	11	10
AB	00	0	0	d	0
	01	0	1	1	0
	11	0	1	d	1
	10	0	d	d	d

$$(5) F = C(\bar{B} + \bar{D})(B + D)$$



2.27有逻辑电路图如图 2.35 所示，

- (1) 写出函数 Y 的逻辑表达式；
- (2) 将函数 Y 化简为最简与或式；
- (3) 用与非门画出其简化后的电路图。

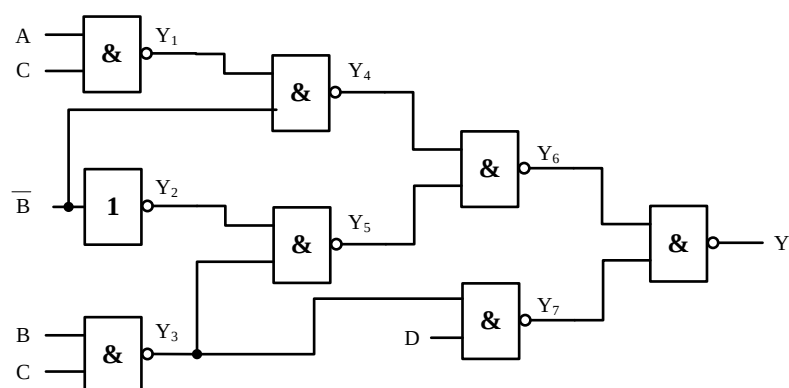


图 错误!文档中没有指定样式的文字。 .1 习题 2.27 电路图

- (1) 逐步写出 Y 的表达式

$$Y_1 = \overline{AC} \quad Y_2 = \overline{\overline{B}} = B \quad Y_3 = \overline{BC} \quad Y_4 = \overline{Y_1 \overline{B}} = \overline{\overline{AC} \overline{B}} \quad Y_5 = \overline{Y_2 Y_3} = \overline{B \overline{BC}}$$

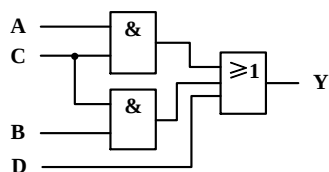
$$Y_6 = \overline{Y_4 Y_5} = \overline{\overline{\overline{AC} \overline{B}} \overline{B \overline{BC}}} \quad Y_7 = \overline{Y_3 D} = \overline{\overline{BC} D} \quad Y = \overline{Y_6 Y_7} = \overline{\overline{\overline{\overline{AC} \overline{B}} \overline{B \overline{BC}}} \overline{\overline{BC} D}}$$

- (2) 化简成最简与或式：

$$Y = \overline{\overline{\overline{\overline{AC} \overline{B}} \overline{B \overline{BC}}} \overline{\overline{BC} D}} = \overline{\overline{AC} \overline{B}} \overline{B \overline{BC}} + \overline{BC} D = (AC + B)(\overline{B} + BC) + \overline{BC} D$$

$$= A\overline{B}C + ABC + BC + \overline{BC} D = A\overline{B}C + BC + D = (A\overline{B} + B)C + D = AC + BC + D$$

- (3) 画出其简化后的电路图。



2.28有逻辑电路图如图 2.36 所示，

- (1) 写出函数 Y 的逻辑表达式；
- (2) 将函数 Y 化简为最简与或式；
- (3) 用与非门画出其简化后的电路图。

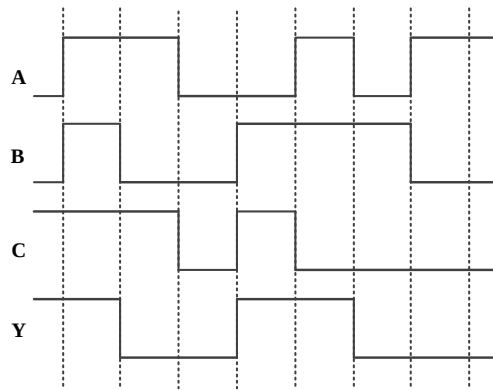


图 错误!文档中没有指定样式的文字。 .2 习题 2.28 波形图

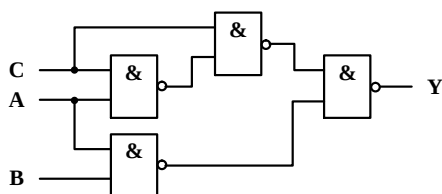
①先写出真值表：

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

②写出 Y 的逻辑表达式，并化简，转化成与非式

$$Y = \bar{A} \bar{B} C + \bar{A} B C + A B \bar{C} + A B C = \bar{A} C + A B = \overline{\overline{\bar{A} C} \cdot \overline{A B}} = \overline{\overline{\bar{A}} \cdot \overline{C} \cdot \overline{A} \cdot \overline{B}}$$

③画出电路图：



2.30 分析【例 1.7】所述事件，Tom 正在野炊，如果下雨或者有蚂蚁，Tom 将不能享受野炊。

(1) 请在对该问题进行逻辑抽象与编码的基础上，分别用真值表、逻辑函数表达式和逻辑电路描述事件。

(2) Tom 要在没有蚂蚁和蜜蜂的晴天去野炊；如果他能听到百灵鸟的歌唱，即使是有蚂蚁和蜜蜂，他也要去野炊。请在逻辑抽象与定义的基础上，分别用真值表、逻辑函数表达式和逻辑电路描述事件。

(1)

①逻辑抽象与编码：

输入变量 1: R 表示有否下雨，R=1 表示下雨，R=0 表示不下雨；

输入变量 2: A 表示有无蚂蚁，A=1 表示有蚂蚁，A=0 表示没有蚂蚁；

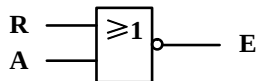
输出变量: E 表示 Tom 是否享受野炊，E=1 表示 Tom 很享受野炊，E=0 表示 Tom 没能享受野炊。

②写出真值表

R	A	E
0	0	1
0	1	0
1	0	0
1	1	0

③写出函数表达式: $E = \overline{R + A}$

④画出电路图:



(2)

① 逻辑抽象：输入变量 4 个：

- A: 有无蚂蚁，A=1 有蚂蚁，A=0 无蚂蚁
- B: 有无蜜蜂，B=1 有蜜蜂，A=0 无蜜蜂
- C: 是否晴天，C=1 晴天，C=0 不是晴天
- D: 有没有百灵鸟唱歌，D=1 有；D=0 没有

输出变量 1 个：定义为 F，=1 表明 Tom 去野炊；=0 表明 Tom 不去野炊

② 列出真值表：

A	B	C	D	F	A	B	C	D	F
0	0	0	0	0	1	0	0	0	0
0	0	0	1	0	1	0	0	1	0
0	0	1	0	1	1	0	1	0	0
0	0	1	1	1	1	0	1	1	1
0	1	0	0	0	1	1	0	0	0
0	1	0	1	0	1	1	0	1	0

0	1	1	0	0	1	1	1	0	0
0	1	1	1	1	1	1	1	1	1

③ 化简：

AB \ CD	00	01	11	10
00	0	0	1	1
01	0	0	1	0
11	0	0	1	0
10	0	0	1	0

$$F = CD + \bar{A}\bar{B}C$$

④ 画出电路图：

